a

Closed-loop optimisation trajectory and application-directed performance of multivariable REBCO films

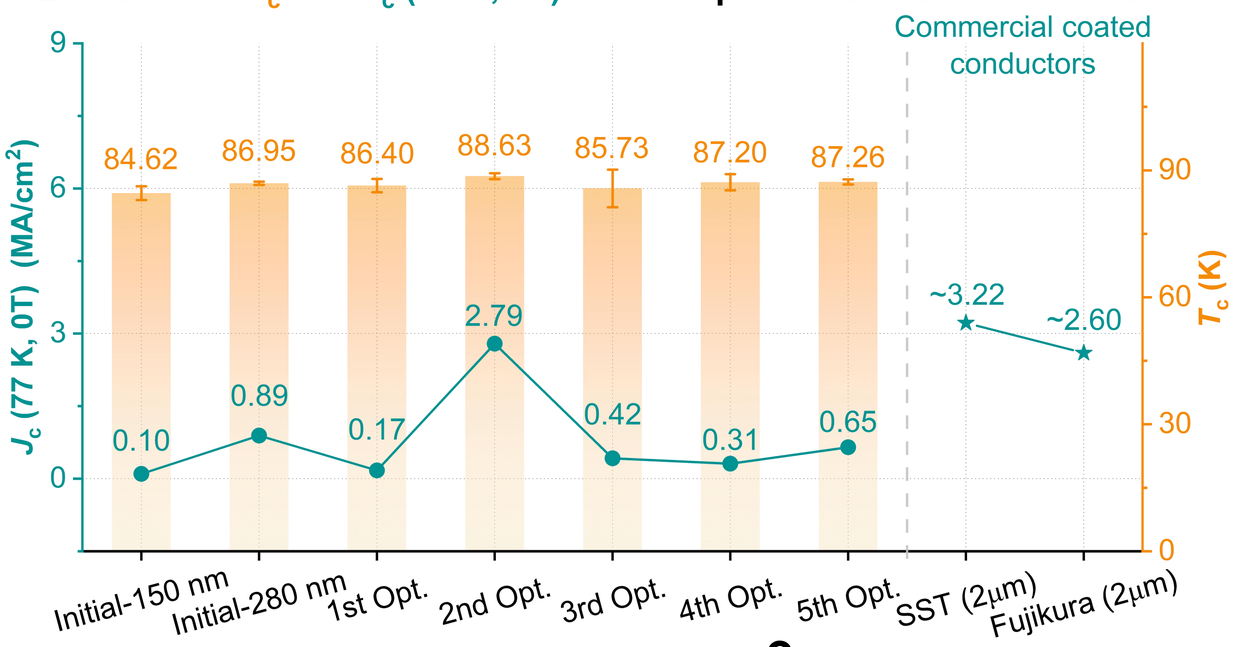
Mechanism-informed closed-loop optimisation trajectory

SupraMAS decision loop: experiment feedback → mechanism diagnosis → parameter update

	1	2	3	4	5	6	7
	Initial-150 nm baseline	Initial-280 nm thickness increase	1st Opt. T_c-oriented search	2nd Opt. densification window	3rd Opt. high-field pining	4th Opt. multi-objective balance	5th Opt. refined optimum
Experimental feedback	Low T_c weak J_c	Improved performance	Performance drop	Best 77K self-field	Strong low-T high-field	Balanced T_c and J_c	Best 4.2 K at 5T
Mechanistic diagnosis	Thin film poor oxygenation	Connectivity enhanced	Low-pressure weak links	Densification window	Vacancies nanoprecipitates	Pinning-oxygen balance	Stable pining optimum
Parameter update	Increase thickness	BO-guided update	High-pressure annealing	Explore low- pressure/high -T	Multi-objective search	Refine annealing	Final recipe

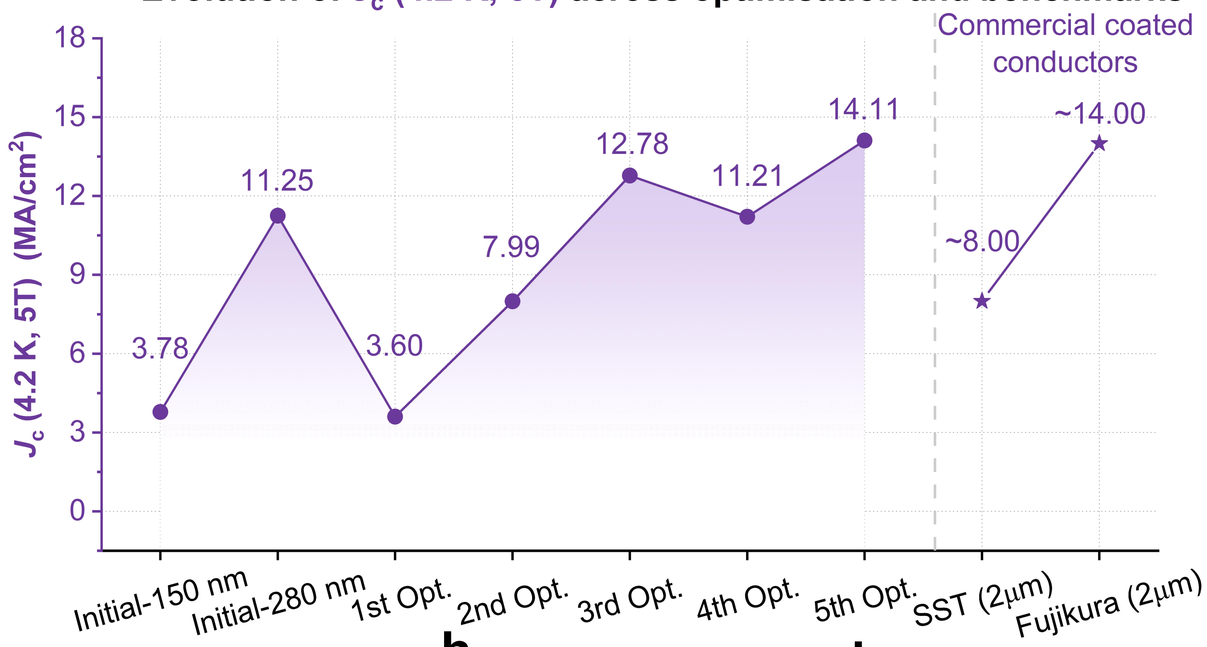
b

Evolution of T_c and J_c (77 K, 0T) across optimisation and benchmarks

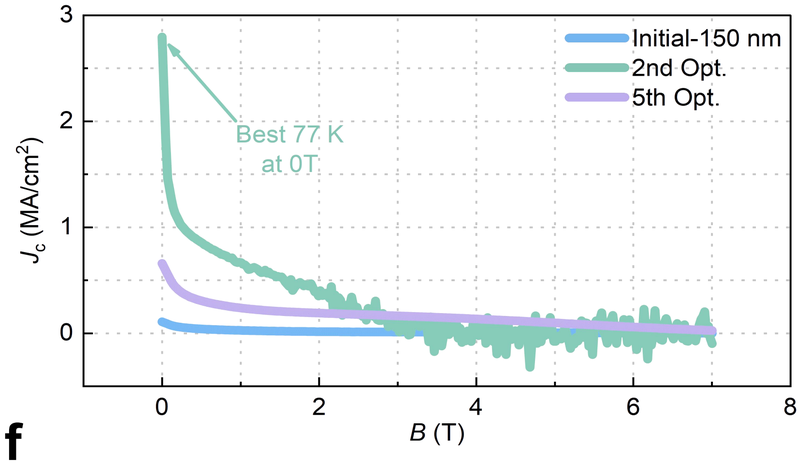


c

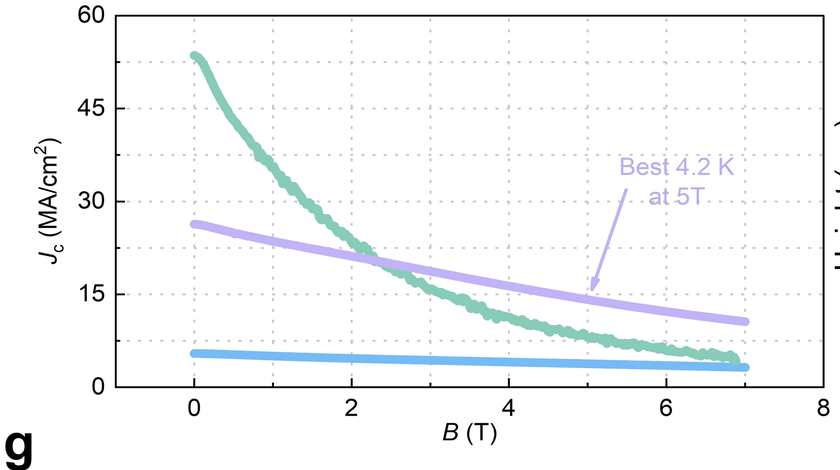
Evolution of J_c (4.2 K, 5T) across optimisation and benchmarks



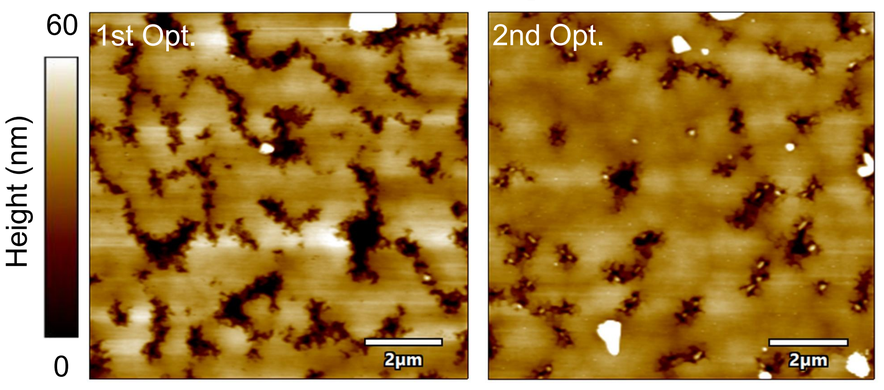
d



e

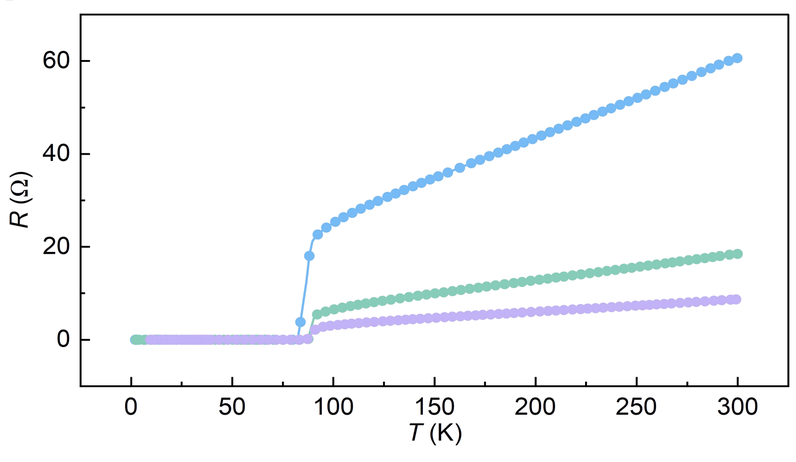


h

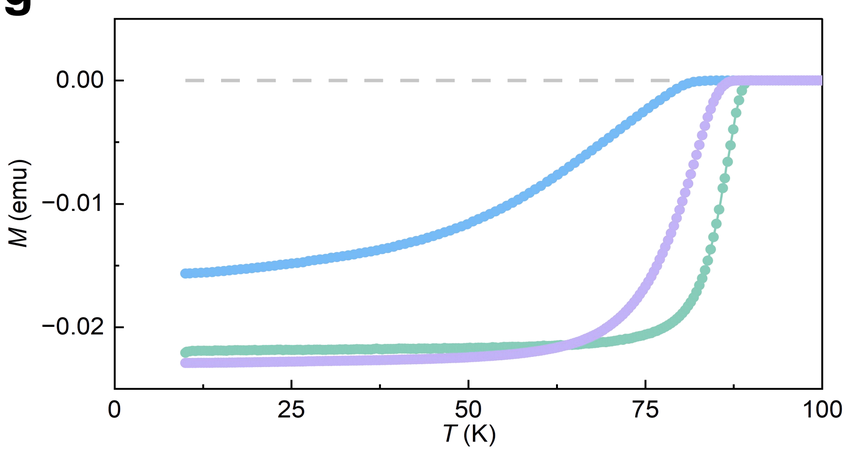


i

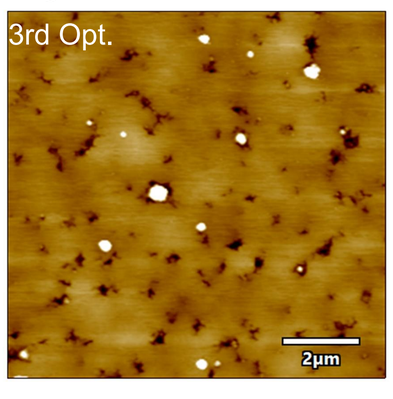
f



g



j



k

